

Cluster Probing: Validation and Stability (k=4)

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1 Same-design label agreement

Table 1: Same-design (higher = better)

Design	Agreement
np	0.778
0p	1.000
2p	1.000
7p	1.000

Represents the ratio of same-design pairs that share a label in each of the 81 unique initializations. For members of the same design, there should be near complete agreement in an ideal scenario.

2 Cross-design label agreement

Table 2: Cross-design (lower = better)

Designs	Agreement
0p vs 2p	0.086
0p vs 7p	0.000
0p vs np	0.165
2p vs 0p	0.086
2p vs 7p	0.000
2p vs np	0.000
7p vs 0p	0.000
7p vs 2p	0.000
7p vs np	0.000
np vs 0p	0.041
np vs 2p	0.000
np vs 7p	0.000

Represents the ratio of cross-design pairs that share a common label in each of the 81 unique initializations. For experiments in different designs, there should be very little overlap in the labeling of the dominant cluster, so they should have low similarity, if any, in an ideal situation.

3 Discussion/Interpretation

From Table 1, 7p same = 1.000, and from Table 2, 7p cross = 0.000 among all other designs. This indicates that the time series from the 7p experiments consistently end up near the same centroid, and that centroid is never shared with other designs.

From Table 1, 2p same = 1.000, and from Table 2, 2p cross = 0.086 only against the 0p design. So there is some overlap in cluster composition between 2p and 0p (which makes sense practically, as the first several μ s of the np designs overlap with the rest of the designs as well), but it is very minor, and the 2p design is otherwise very self-consistent across all 81 experiments.

From Table 1, 0p same = 1.000, and from Table 2, 0p cross = 0.086 against 2p and 0.165 against np. This means that the 0p design has good internal consistency and is well-separated from the 7p design, but also has some overlap with the other two designs. This design has the weakest separations of the whole set, although the regions of overlap are very low and are otherwise relatively consistent.

From Table 1, np same = 0.778, and from Table 2, np cross = 0.165 against 0p. The lower same-design agreement means that about 22% of same-design pairs disagreed on cluster label among the 81 experiments. However, since the cross-design agreements are all relatively close to zero, the np design is likely sitting near a cluster boundary. Having seen plots of the time series from multiple designs overlaid, I've noticed that the np time series tends to overlap with all other designs for the first several μ s, so it makes sense that it would be relatively confused or inconsistent within itself.

Over all, the same- and cross-design analysis seems to point towards all 81 training initializations resulting in fairly consistent clustering, with the same common themes that 7p clusters exceptionally well, 2p and 0p are distinct from the other designs but not necessarily from each other, and np is mostly distinct and always has its own exclusive cluster, but sometimes has edge points from its data end up in other clusters than its dominant one.